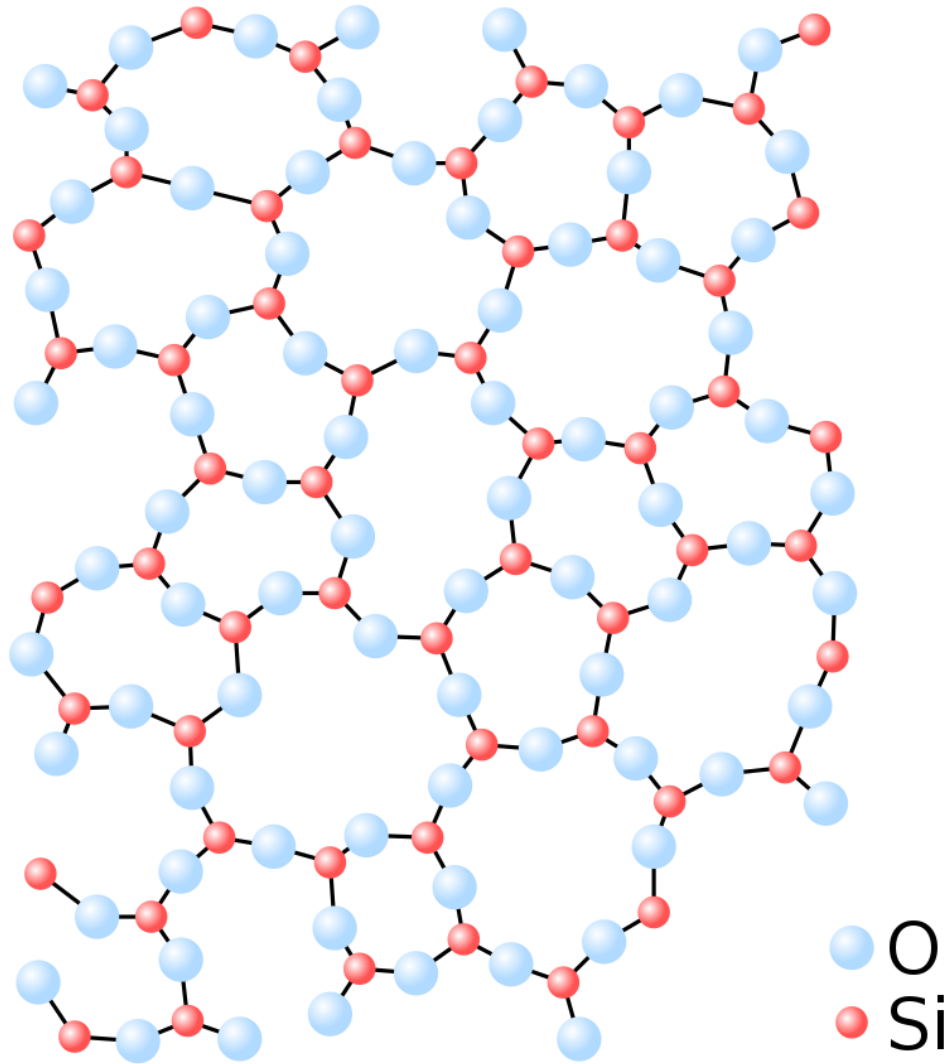
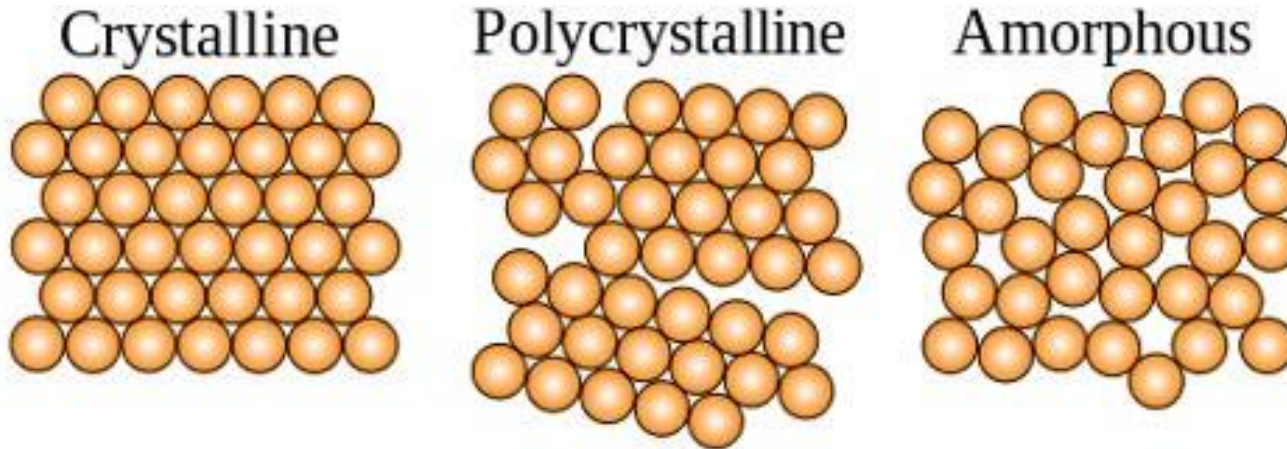


Chapter 1 Structure of Glass



Atomic Structure of Solids



GLASSES - subset of amorphous solids

1. A glass is an inorganic product of fusion which has cooled to a rigid condition without crystallisation
2. A glass is a material which exhibits the glass transition phenomenon

Better: A glass is a solid that possesses no long-range atomic order and which undergoes the glass transformation from solid to supercooled liquid on heating.

(Not all amorphous solids exhibit the glass transition – these are not glasses)

Solid-like	- rigid, hard, fractures (rapid processes)
Liquid-like	- flows (high T or long t) (slow processes)
	- structure

Glass Formation

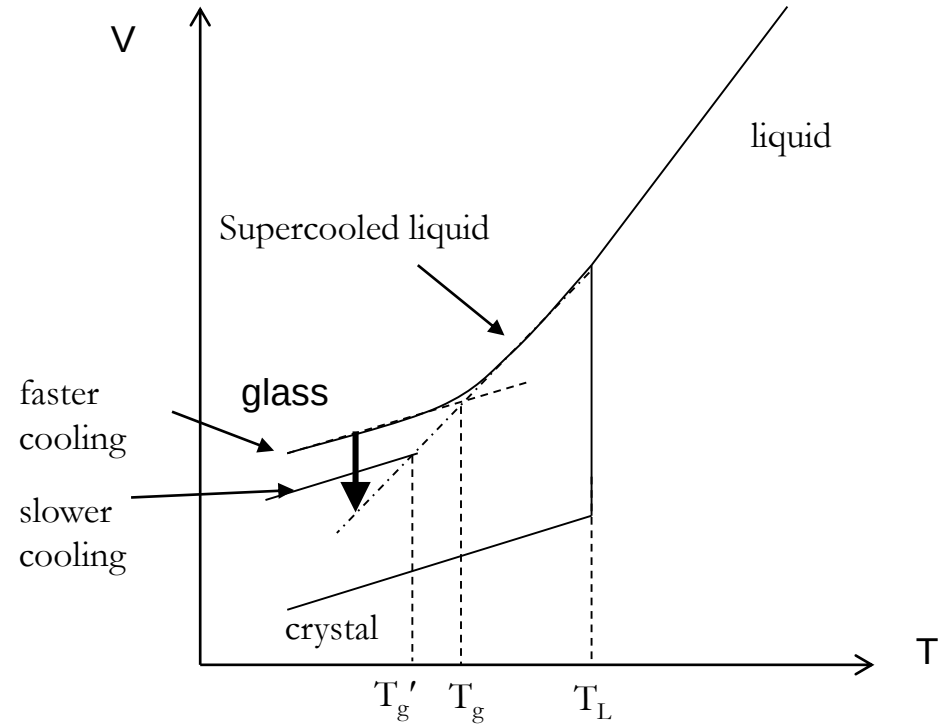
Cooling liquid

T_L - liquidus/crystallisation temperature
Most liquids crystallise below T_L with change in volume;

Some continue as supercooled liquids
Depends on structure and cooling rate

dV/dT of supercooled liquid same as liquid until T_g

T_g – glass transition temperature
- below T_g , supercooled liquid behaves like solid - GLASS



Value of T_g depends on cooling rate

If glass held at a temperature near T_g , V will decrease towards equilibrium value which supercooled liquid would have had (arrow) – v. slow.

Supercooling

- ✦ Possible because of v. high E_a for phase transformation from liquid/glass to crystal
- ✦ T_g important parameter in technology
 - lower limit of use of rubber
 - upper limit of use of thermoplastics
 - annealing – stress relief – just below T_g
- ✦ Liquid structure closer to glass than crystal $\therefore E_a$ for transformation from liquid $\rightarrow$ glass less than for liquid $\rightarrow$ crystal
- ✦ E_a depends on strength and directionality of bonds in glass $\therefore$ electron-pair > ionic > metallic
 - metallic glasses - need cooling rates of $> 10^6 \text{ K s}^{-1}$
 - soda-lime-silica - “ “ “ “ $\sim 10 \text{ K s}^{-1}$
 - SiO_2 - “ “ “ “ $\sim 0.1 \text{ K s}^{-1}$
 - B_2O_3 - will **not** crystallise at normal pressures

Stability of Glass

- there are NO conditions under which glasses are thermodynamically stable

-thermodynamically metastable



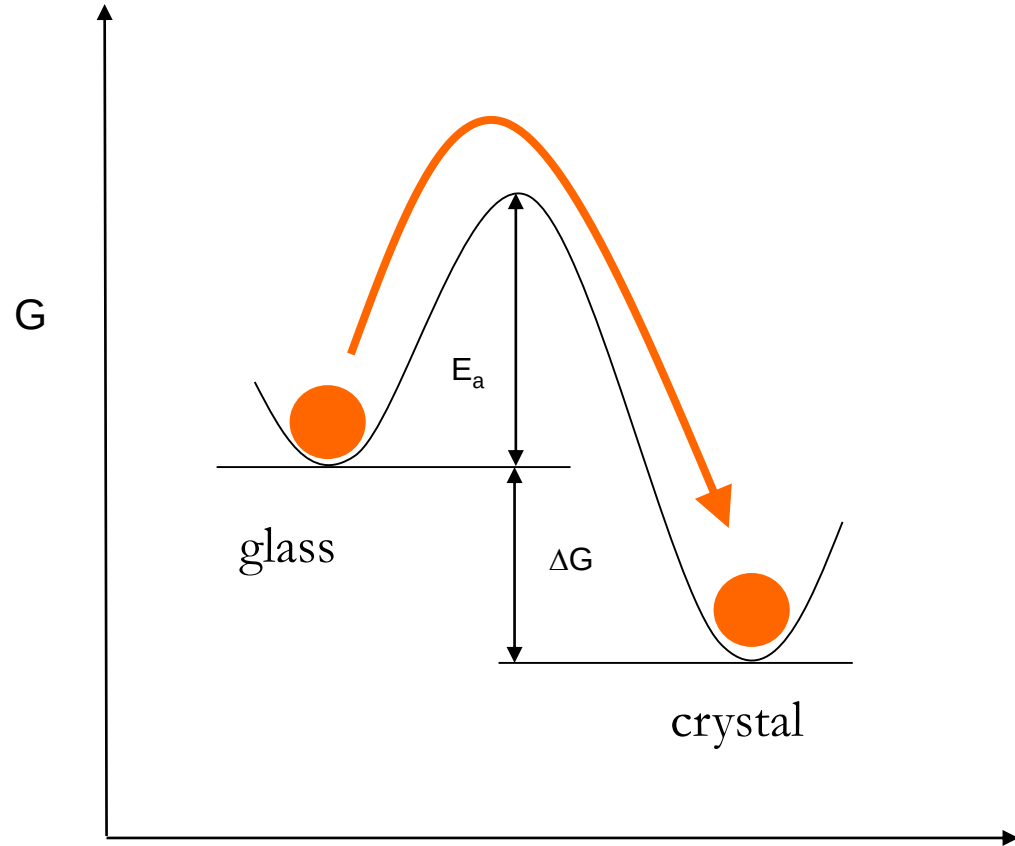
Stable – c.o.g. at lowest point



Unstable – c.o.g. already at highest point – any change will move it to stable position



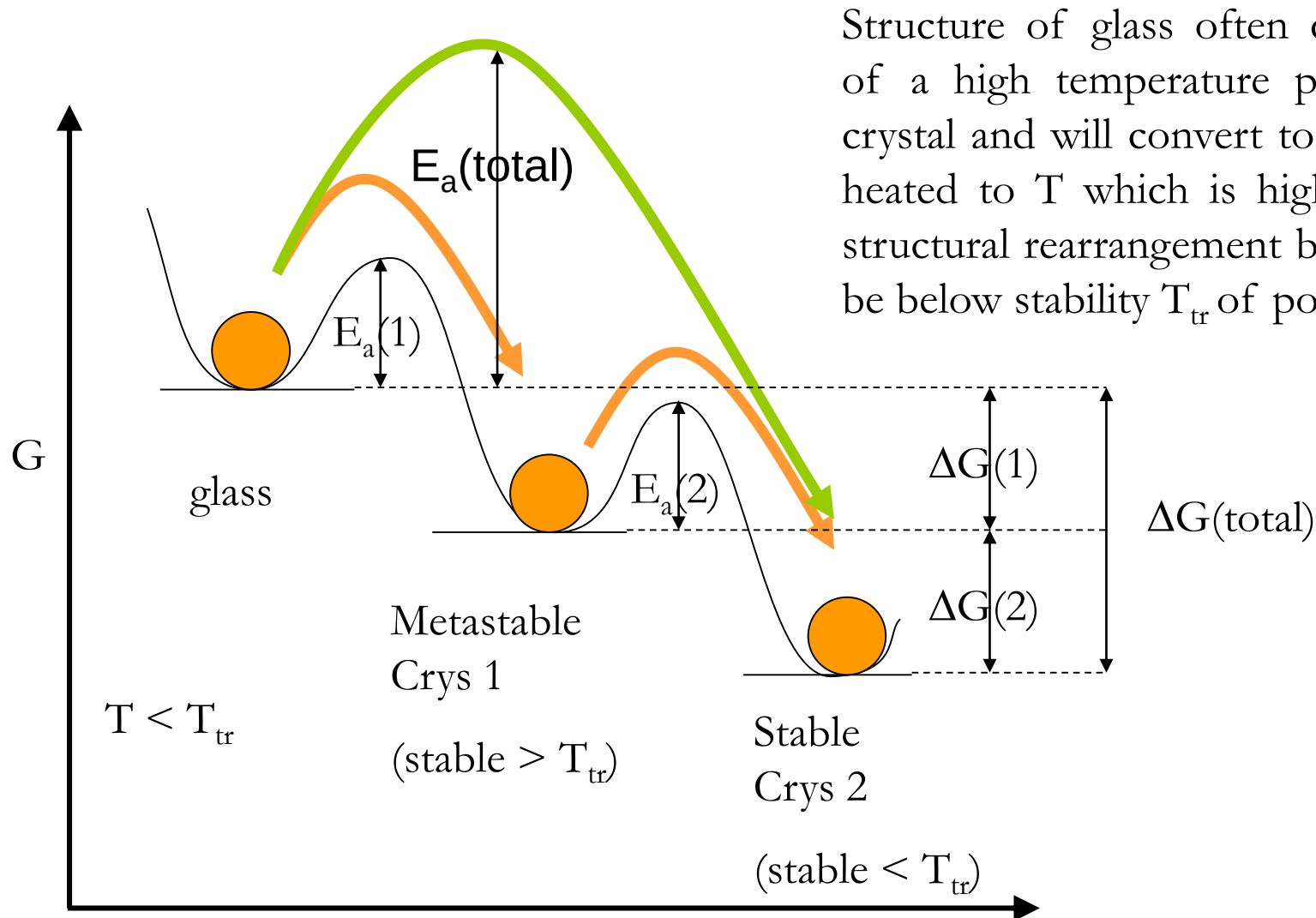
Metastable – c.o.g. has to rise before it falls



Glass at higher G than crystal but needs to acquire activation energy E_a before it can convert by rearranging structure.

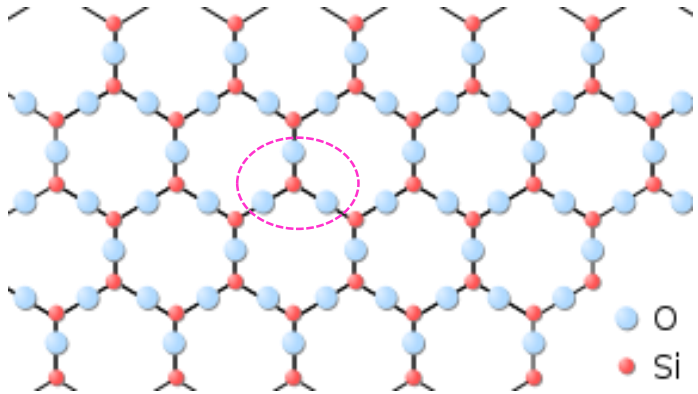
Crystallisation

If the glass is given enough thermal energy, bonds can break and rearrange to a crystal phase but this may not be the expected phase

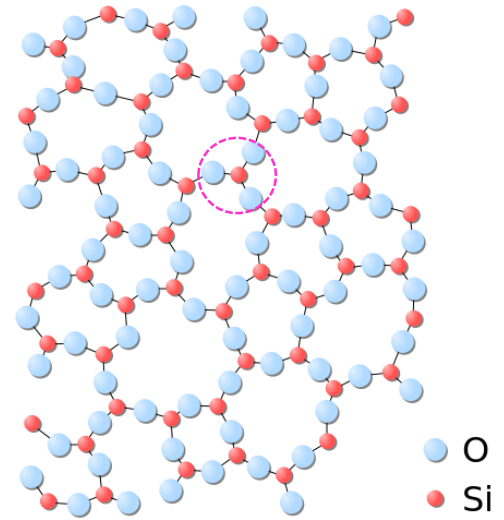


Structure of glass often closer to that of a high temperature polymorph of crystal and will convert to that phase if heated to T which is high enough for structural rearrangement but which may be below stability T_{tr} of polymorph.

Glass Structure

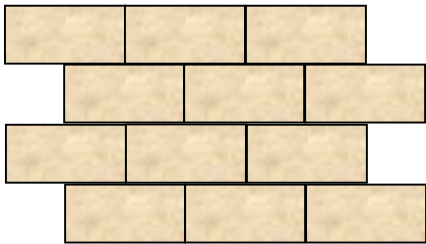


- Range of M-O-M bond angles
- Range of ring sizes

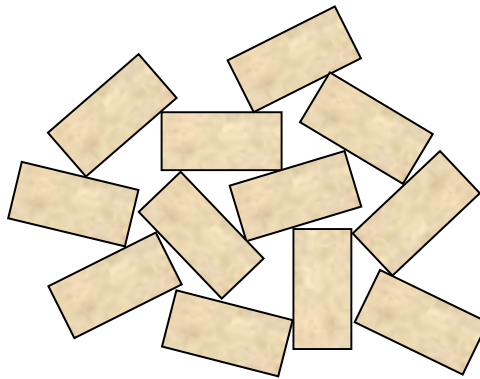


The periodic crystalline lattice structure of SiO_2 in two-dimensions

The random network structure of glassy SiO_2 in two-dimensions.

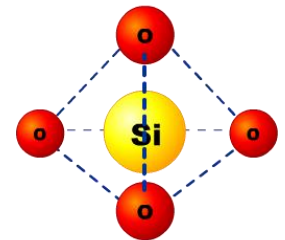


Crystal: long-range order



Glass: short-range order

Same structural units



Models of Glass Structure

Random Network Theory (Zachariasen 1932)

Empirical rules derived from observation of oxide glasses A_nO_y

1. O not coordinated to more than two atoms of A
2. Coordination of A small (Si 4; B 3 or 4)
3. coordination polyhedra of A must share corners not edges or faces
4. at least 3 corners of each polyhedron must be shared to give a 3D network and at least 2 corners must be shared for a glass to form

Implies 1st coordination sphere about A is well-defined and nearly identical to crystal (SHORT RANGE ORDER - SRO)

Most common coordination polyhedron found in oxide glasses is the tetrahedron (rule 2) e.g. $[\text{SiO}_4]$, $[\text{GeO}_4]$, $[\text{PO}_4]$, $[\text{AlO}_4]$, $[\text{BO}_4]$,

Octahedron – edge and face sharing breaks rules 1, 2 and 3.

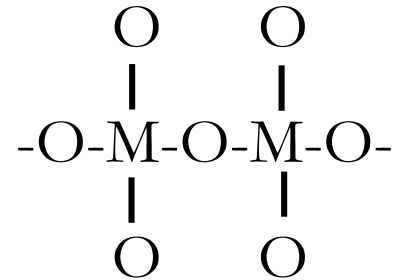
Role of Different Oxides in Glass Structure

Network formers

- e.g. SiO_2 , B_2O_3

strong, directional e-pair bonds

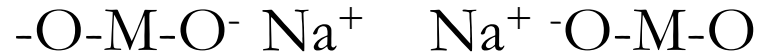
- can form a glass network alone



Network modifiers

- e.g. Na_2O , CaO

- break the M-O-M links in the glass



ionic bonds

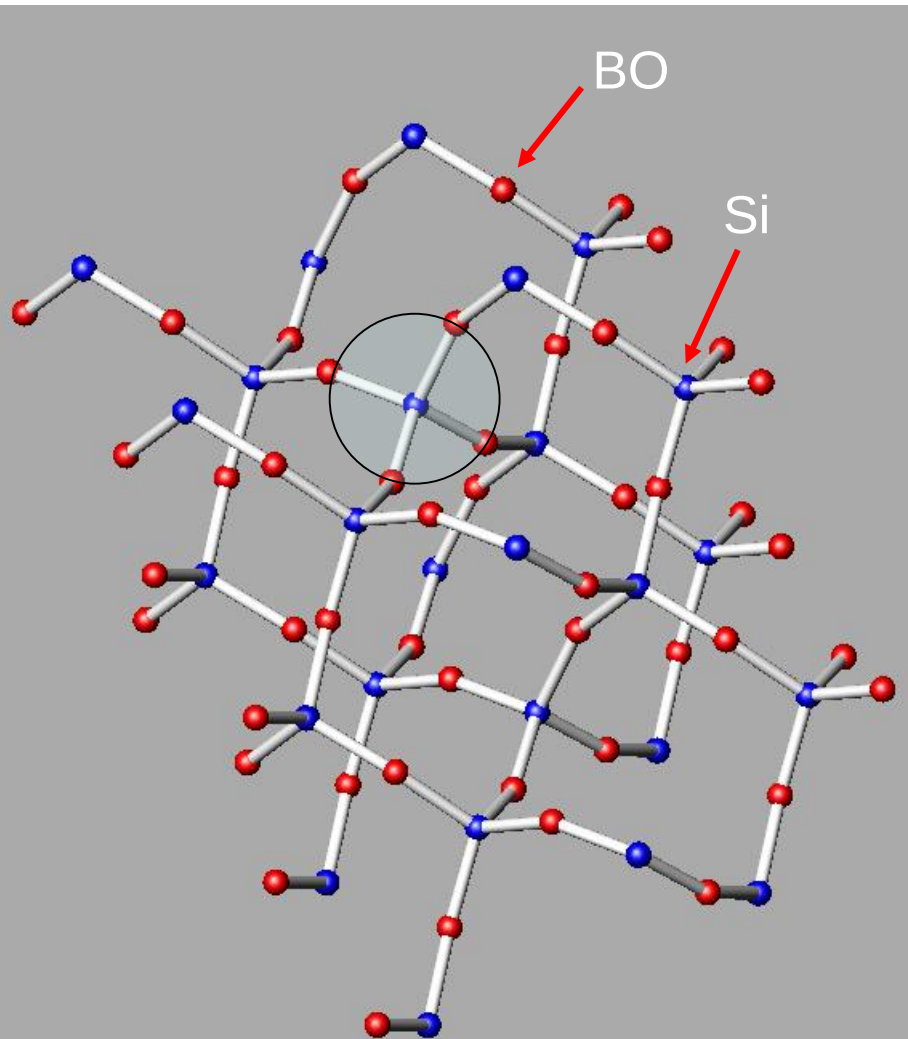
Network intermediates

- e.g. Al_2O_3

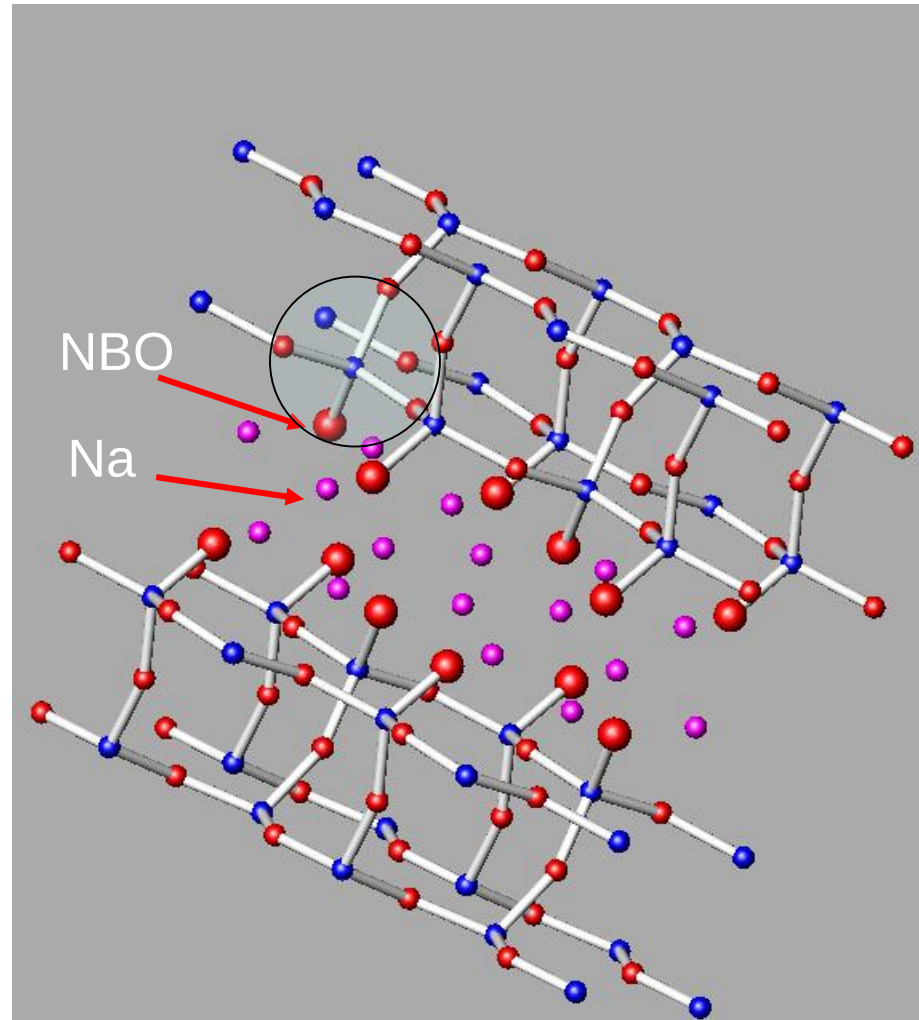
- repair the breaks



Random Network Model

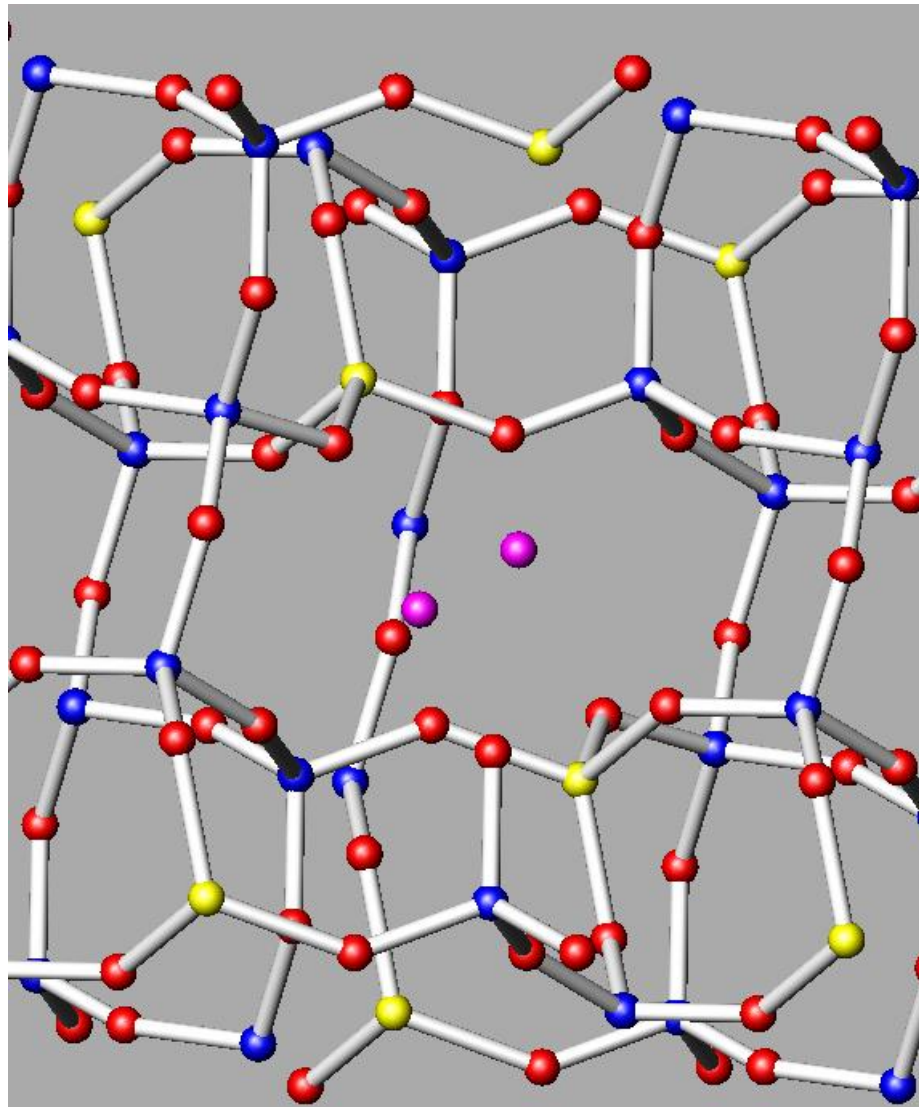


Network former SiO_2
all bridging oxygens (BO)



Add network modifier Na_2O
form non-bridging oxygens (NBO)

Random Network Model

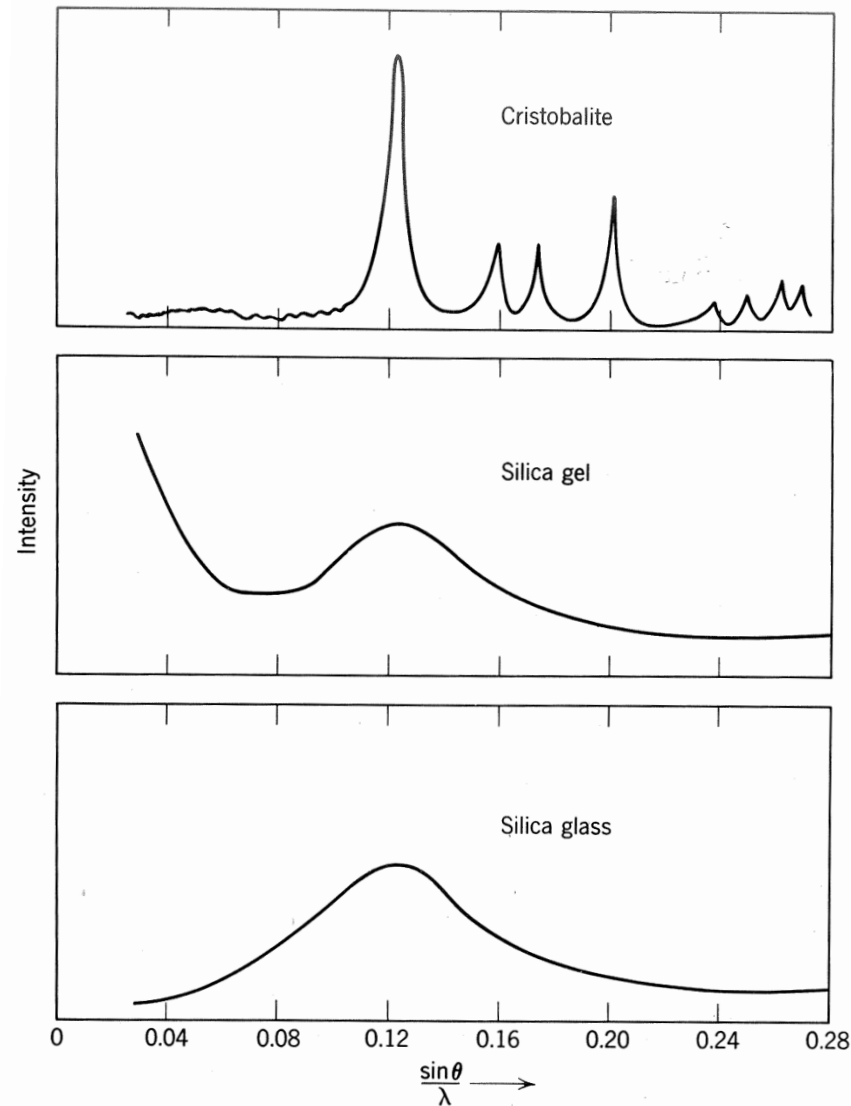


form $[\text{AlO}_4]^- \text{Na}^+$
groups with all BO

If $\text{Na}_2\text{O} = \text{Al}_2\text{O}_3$
all NBO removed

Add intermediate oxide Al_2O_3
remove non-bridging oxygens (NBO)

X-Ray diffraction Patterns Difference



Types of Ceramic Glasses_Silicate glass

1. Fused silica glass

SiO_2 , has very low thermal expansion, very hard and resists high temperatures (1000–1500°C), the most resistant against weathering.

Applications: furnace tubes, melting crucibles, etc



2. Soda-lime-silica glass (window glass)

silica 72% + sodium oxide (Na_2O) 14.2% + lime (CaO) 10.0% + magnesia (MgO) 2.5% + alumina (Al_2O_3) 0.6%.

Transparent, easily formed, most suitable for window glass, a high thermal expansion, poor resistance to heat (500–600 °C).

Applications: Windows, containers, light bulbs, tableware.



Types of Ceramic Glasses

3. Sodium borosilicate glass, Pyrex

silica 81% + boric oxide (B_2O_3) 12% + soda (Na_2O) 4.5% + alumina (Al_2O_3) 2.0%.
Stands heat expansion much better than window glass.

Applications: chemical glassware, cooking glass, car head lamps, etc.



4. Lead-oxide glass, crystal glass

silica 59% + lead oxide (PbO) 25% + potassium oxide (K_2O) 12% + soda (Na_2O) 2.0% + zinc oxide (ZnO) 1.5% + alumina 0.4%. Has a high refractive index, making the look of glassware more brilliant (crystal glass).

Other glass: Phosphate glass, Chalcogenide Glass, Fluoride Glass.....